



Distance, Mobility, and Mode Choice: A Spatial Analysis of School Travel Behaviour in Ireland

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Background and Problem

- Daily travel to school significantly influences physical activity levels in children and adolescents
- Increased use of unhealthy transport has reduced walking and cycling
- Active travel provides important opportunities for regular physical activity
- In Ireland, school location, distance to school, and neighbourhood features affect travel choices
- Travel habits formed in childhood often continue into adulthood

Objectives of Project

1. Map and analyse the spatial distribution of schools across Ireland.
2. Examine the distance between students’ residential locations and their schools, and investigate student mobility related to school attendance.
3. Analyse school travel behaviour and assess the prevalence of healthy versus less healthy commuting modes.
4. Explore the relationship between school proximity, travel mode, and potential health implications.

Data Sources and Datasets

Data on schools in Ireland are of the following Types:

-School-Level Data (Department of Education): School type, enrolment, and geographic location (Eircode,latitude, longitude). -Settlement-Level Travel Data (CSO): Census data on modes of travel to education at settlement level across multiple years.

-Spatial Boundary Data (CSO): Geographic boundaries for towns and urban areas used for spatial mapping and analysis.

Early Results

As a first step in the analysis, we focused on understanding the data we are working with. To achieve this, we explored the spatial distribution of schools across Ireland. By mapping school locations and producing a density plot, we were able to visualise areas with high and low concentrations of schools.

This initial exploratory analysis helps to validate the dataset, identify spatial patterns, and highlight regional differences in school availability. Understanding where schools are clustered provides an essential foundation for subsequent analyses on school accessibility, travel distances, and the use of healthy versus unhealthy modes of transport to school.

Overall, the graph highlights a strong urban–rural divide in the spatial distribution of schools. Identifying Dublin as the main density hotspot is important, as such concentration is likely associated with shorter travel distances and a greater potential for active modes of transport, while lower-density regions may rely more heavily on motorised transport options due to longer travel distances.

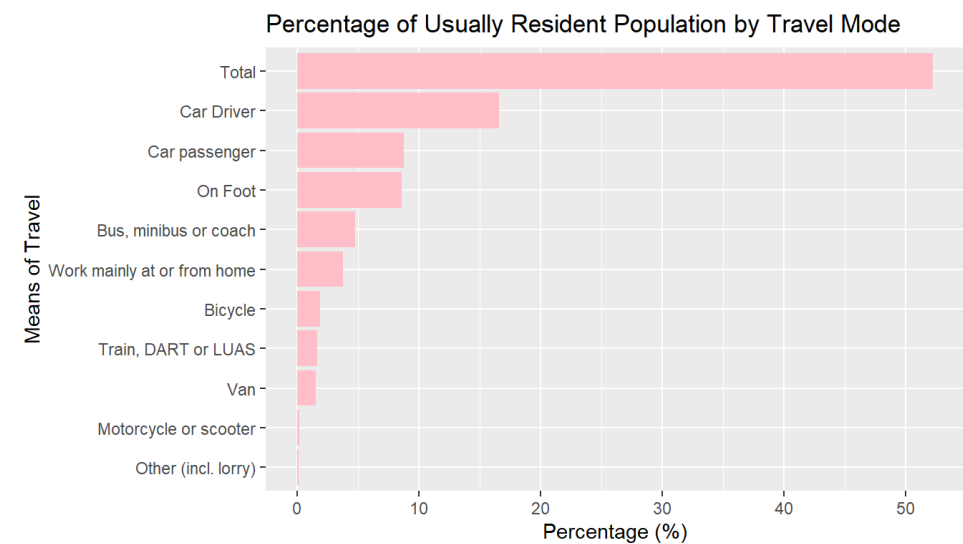
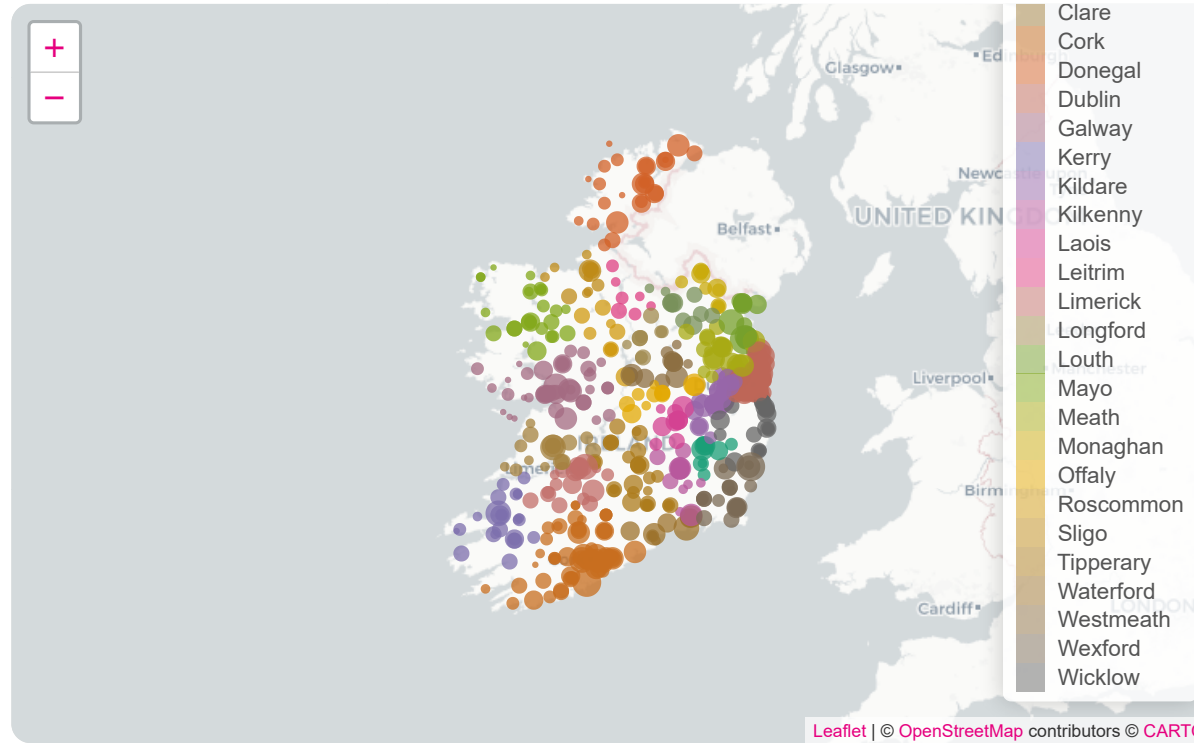


Figure 1: Great figure!



Kriging is a geostatistical interpolation method that uses spatial dependence to predict values at unsampled locations, making it useful for identifying spatial patterns and inequalities related to school locations, travel behaviour, and potential health impacts.

[using ordinary kriging]

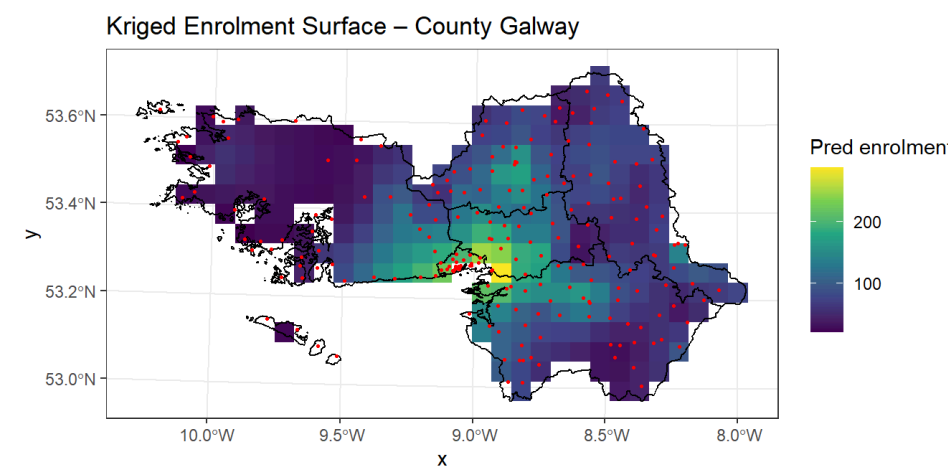


Figure 2: Kriging schools in Co Galway

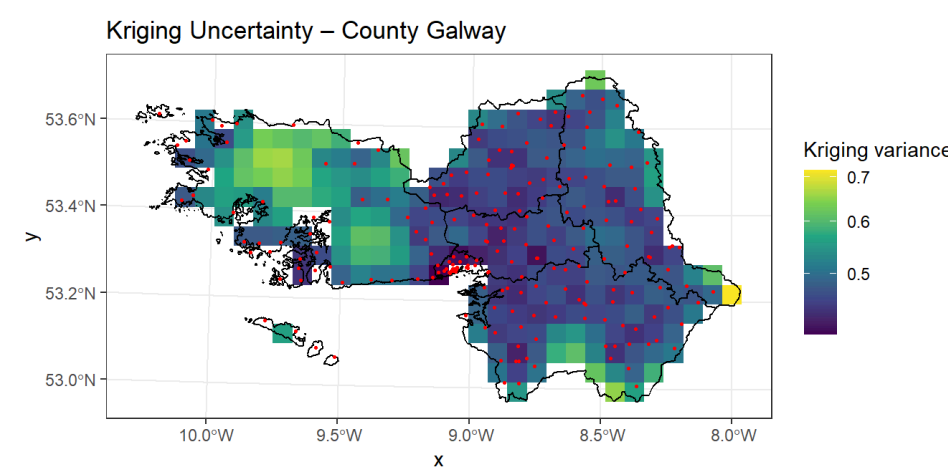


Figure 3: Kriging schools in Co Galway

The kriged surface shows higher predicted enrolment in areas where schools (red points) are more densely

Next Project Steps

-Define outcome variables (e.g. enrolment size, accessibility, resource availability) -Control for confounding factors (socioeconomic context, location, school size, demographics) -Apply county-level kriging to examine spatial variation in school enrolment -Extend analysis using regression or multivariable models to assess relationships between distance, travel mode, and school characteristics

GitHub

This project can be viewed at our GitHub repository here: <https://github.com/judithbeltranlopez/Final-Project>

References